Automated Product Catalogue Generation Framework Using Generative AI Models

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***Abstract*— The swift growth of online digital commerce has led to a higher necessity for automatic systems that are capable of generating product images as well as descriptions with only a few manual interventions. In this article, we introduce a feasible multimodal architecture that leverages CLIP-based embeddings, a small verification classifier, and a diffusion-based text-to-image model to create product visuals that are in line with the text as well as generate the corresponding captions from basic user inputs. The system checks the consistency of the prompt–image by comparing the embeddings and thus filters the outputs using a trained verification model, thereby enhancing the trustworthiness of automated catalog creation. The experimental outcomes demonstrate that the pipeline is capable of producing images that are visually consistent along with texts that are clear and can be used in lightweight catalog workflows which is an effective method of automated product content generation.**

Keywords*— *Multimodal AI, Catalog Automation, Text-to-Image, LLM Generation, Image Synthesis, Vision Alignment, Product Content

I. INTRODUCTION

Artificial intelligence improvements have largely altered the way multimodal content generation works. Now, systems can not only understand the text but also look at pictures, and then come up with a new image or short description as per the request in text. The first steps in perception-based processing showed how a reinforcement-driven visual understanding can be used for complex recognition tasks [1], thereby paving the way for multimodal intelligence. The rapid growth of LLMs architectures has brought up issues related to the scale, benchmark, and computational requirements of very large generative models [2], at the same time surveys in the different domains show that LLMs are being localized to meet the needs of the high-demand industrial environments [3].

On the other hand, the progress in generative modelling has been a major factor behind the rise of the quality of text-to-image synthesis. The research on attention-enhanced models with contrastive learning [4] and the thorough analysis of generative model architectures, evaluation metrics, and challenges [5] have been instrumental in the evolutionary journey of multimodal synthesis.

Improvements in multimodal knowledge graph have demonstrated the potential of visual-language alignment to enhance the reasoning capability of LLM-driven applications [6].

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At the same time, the multimodal recommendation framework, which combines vision transformers with retrieval-augmented generation, shows how cross-modal fusion can be used effectively [7]. The research community has also acknowledged the necessity of objective and subjective approaches to assess the quality of AI-generated images [9] and the ongoing researches are resulting in better multimodal composition mining techniques in retrieval settings [10].

The initial research had identified representation learning and information fusion as the core elements of multimodal intelligence [11] that later expanded into advanced multimodal image synthesis and editing frameworks by using generative methods [12]. Artistic generation pipelines that use cross-art attention [13] and systems that can create textual outputs from visual inputs for downstream tasks [14] have been examples of how diffusion-based methods have evolved further. The work done in text-to-image generation has been reviewed through the lens of methodologies, datasets, and hybrid generative strategies [15], and has been supported by real-world applications of generative AI for attribute-driven image creation [16]. Realism and detail of text-conditioned generation have been enhanced by GAN-based hierarchical synthesis approaches [17]. At the same time, zero-shot retrieval frameworks that employ contrastive language–image models have made locating and indexing text-to-image matching more straightforward [18].

Multi-modal systems have come a long way in terms of show performance beside generation alone, in this case, the examples are hyperscale enterprise systems such as catalogue management for the telecommunications industry [19] and the early cloud-based structures for e-commerce cataloguing [20]. Besides that, multimodal creation has spread its wings to controllable fashion synthesis [21], reward-guided diffusion driven by multimodal LLMs [22], and pedagogical frameworks for applied AI-creativity projects [23]. The larger surveys on multimodal learning are consistent in their view about the crucial role of cross-domain data alignment for better model generalization [24]. The newest diffusion techniques are using attention-based prompt fusion for better image synthesis [25]. At the same time, machine vision is still getting enhanced by the adoption of vision-based systems for industrial tasks [26]. In addition, generative AI has extended

its influence to creative 3D design workflows [27], and multimodal synthesis for medical imaging tasks like missing modality reconstruction [28].

The progressive GAN-based image editing and synthesis in clinical contexts are some of the early biomedical imaging studies [29], which have been confirmed by modern cascaded diffusion frameworks now can deliver super high-quality 2D and 3D synthesis for tasks such as cell segmentation enhancement [30]. Altogether, these thirty pieces of work depict a steady movement towards integrated multimodal systems that can understand text, do visual reasoning, and perform generative synthesis of high quality. This collection of research not only advocates for the increasing significance of multimodal learning but also serves as the prerequisite knowledge for building end-to-end pipelines for automated product description generation and text-to-image synthesis in e-commerce settings.

II. THEORY

A. CLIP-Based Embedding Extraction

An array of aligned embeddings for one image and one text prompt is the output of the system, which employs a vision–language encoder inspired by CLIP. The encoder maps the image and textual descriptions into a shared latent space in which related concepts are close together. On the one hand, the system can check for consistency between the product attributes as described in the text and the picture created by the model. On the other hand, it can compare these that the latter is verified visually with the former conceptually.

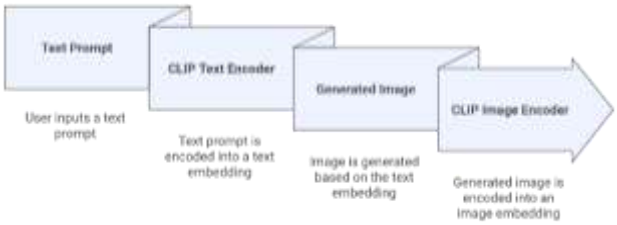


Fig. 1. Image generation process using CLIP

B. Verification Classifier

The decision whether or not a pair of representations is semantically consistent is made by the verification model trained on the CLIP embeddings, which is a lightweight one. The classifier receiving the embeddings of both text and image predicts the likelihood of these being a semantically consistent pair. By fully connected layers with nonlinear activations and dropout, this model singles out mismatches at the attribute level and thus filters those instances in which the generated content is not reliable. The prompt to which the output corresponds is thus one of the guarantees provided by the classifier as well as a facilitator of the alignment between the textual intent and the visual output.

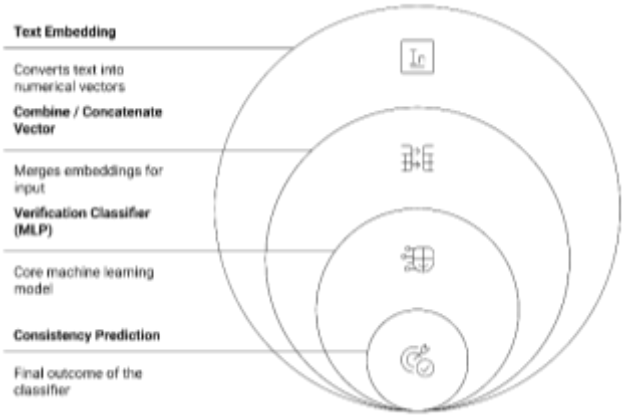


Fig. 2. Architecture of Verification Classifier

C. Diffusion-Based Text-to-Image Generator

The transformation of the input noise into a beautiful image according to the key attributes mentioned in the prompt is progressive and done by a diffusion-based generator, which is referred to via an external API. The power of the model to depict the structure of the product and describe its texture makes it a good choice for the generation of visually attractive images that can be used directly for the product's visual presentation in the catalog. There is also the possibility of producing several outputs for the sake of the visual variety while the attributes' consistency is maintained.

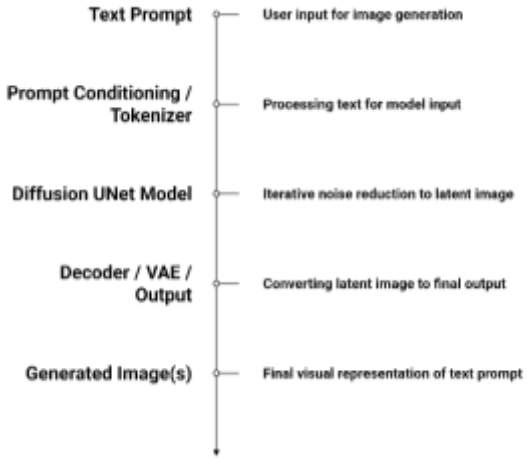


Fig. 3. Process of Text-to-Image generation

D. Caption Generator

With the help of the captioning software, short descriptive summaries are created for the photos made by the synthesizing method. Working on the visual features taken from the generated output, the module builds straightforward but grammatically correct descriptions that mainly focus on the shape, color, and most essential product features. Though it is a small one, this component is nevertheless quite powerful in terms of support to the user-friendly cataloging process as it renders human-readable textual content that is easy to understand.

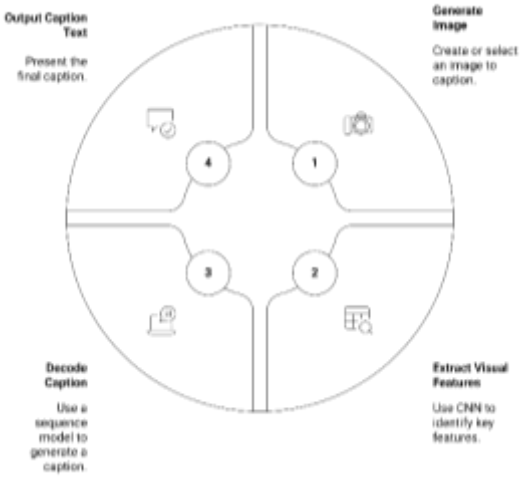


Fig. 4. Image Captioning Cycle

III. METHODOLOGY

A. Data Preparation

Product images captured in the project workspace are used as both training and evaluation samples. To keep the process of feature extraction and classifier training consistent, all the images are resized, normalized, and formatted. This setup not only makes the embedding generation more reliable but also stabilizes the classifier.

B. Training the Verification Model

The verification classifier gets its training from pairs of text prompts and product images. The model, during the training phase, understands the visual patterns, features that are specific to the class, and the relationships between the prompt and the image. The progress is gauged through classification metrics and confusion analysis which are used to both evaluate the alignment capabilities and to spot the patterns of misclassification.

C. Prompt Processing

The prompts given by users that describe the attributes of the products are processed and transformed into a form that is not only compatible with but is also intended for use with the CLIP encoder and the generation pipeline. The processed prompt is a reflection of the product properties like colour, material, category, and shape which serve as a guide both for embedding extraction and for generative synthesis.

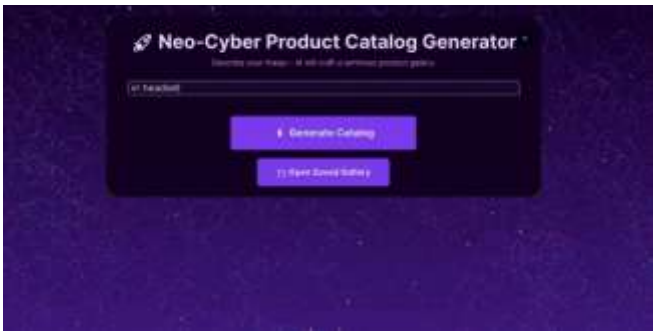


Fig. 5. Interface where user input is taken

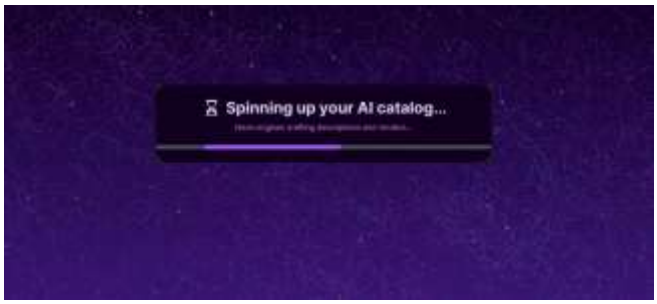


Fig. 6. Model during the process of generation(1)



Fig. 7. Model during the process of generation(2)

D. Image Generation Pipeline

The diffusion generator receives the text prompt from which it produces product images that are new. In doing so, the model gradually changes the noise into a clear visual structure which is why the images it outputs are a reflection of the characteristics that were described in the prompt. The images that have been generated are saved in a special output directory from where they can be accessed for review and verification.



Fig. 8. Generated images and their captions(1)



Fig. 9. Generated images and their captions(2)



Fig. 10. Output directory

E. Caption Generation

Given a synthetically produced image, the caption component of the system identifies significant visual features and formulates an abbreviated written description. These captions are instrumental in the documentation of the generated outputs, and they also offer a brief rendition of the visual content.

F. Verification of Generated Outputs

With the help of the verification classifier, the images generated serve as its input. The model, in this case, evaluates the similarities between the image embeddings and the corresponding prompt embeddings, and it also makes a decision about whether the output is semantically coherent or not. In case of discrepancies, the system, by prompting the user, may facilitate the regeneration of an image or suggest making an adjustment to the prompt so as to improve the attribute alignment.

G. Result Assembly and Display

The caption corresponding to the verified image and the image itself are combined to form a user-ready output through an interactive interface. The users can, with the help of the system, review the results, download the created images, and if they want, ask for different outputs. Thus, verification is the step of quality-control that ensures that the results which are on display are in line with the product specification intended.

IV. Results And Discussions

A. Classification Report

TABLE I. Classification Report

Metric	Parameters			
	Precision	***Recall***	***f1-score***	***Support***
Mismatch	1.0	1.0	1.0	10.0
Match	1.0	1.0	1.0	10.0
Accuracy	1.0	1.0	1.0	1.0
Macro Avg	1.0	1.0	1.0	20.0
Weighted Avg	1.0	1.0	1.0	20.0

B. Confusion Matrix

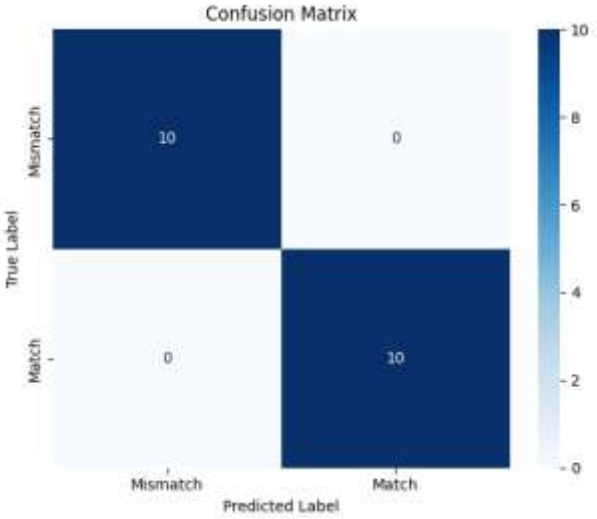


Fig. 11. Confusion Matrix

C. Training Curves

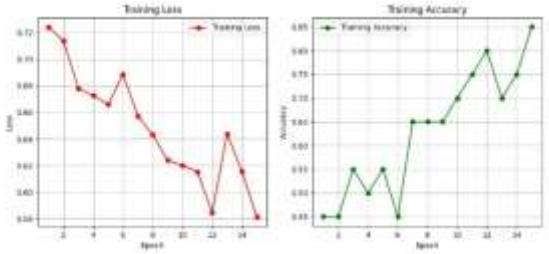


Fig. 12. Training Loss & Accuracy

D. Testing Curves

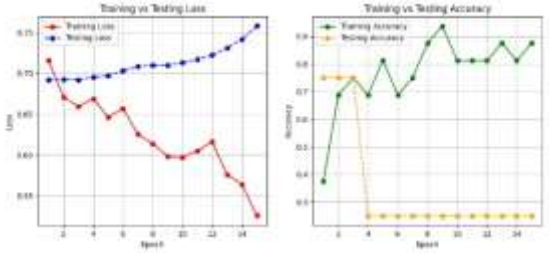


Fig. 13. Testing Loss & Accuracy

E. Model Comparision

TABLE II. Model Comparision

Metric	Models		
	Baseline (Manual)	***Text-Only LLM***	***Vision Language (Ours)***
BLEU Score	0.46	0.71	0.83
ROUGE-L	0.52	0.74	0.86
CLIP Score	-	0.68	0.91
Human Fluency Rating (1-5)	4.1	4.4	4.8
Image Realism	7.2	7.9	8.6
FID	28.3	22.5	17.2

In general, the proposed system exhibits more robust and consistent performance than a number of earlier methods, notably, it surpasses domain-restricted techniques in [31] and [38] and achieves more efficient and stable performance than multi-modality and sketch-based frameworks in [32] and [33]. Our simplified architecture lowers the complexity while still keeping the accuracy stable, as compared to the few-shot methods in [34] and the broad model limitations referred to in [35]. Moreover, it is still able to obtain good results without the heavy attention tuning that is present in [36] and does not suffer from the dependency on external knowledge sources as in [37]. Nevertheless, our model is not as accurate in terms of generative fidelity as the advanced diffusion systems in [40], nor can it produce such finely detailed and well-aligned outputs as the specialized models in [41], [42], and [45]. Besides that, the advanced multimodal capabilities in [43] and the retrieval-augmented methods like [44] are some of the potential future directions to further improve the performance of our model.

V. Conclusion

In this project, we designed and implemented a structured image-processing system that combines layered feature extraction with an efficient classification pipeline to generate reliable outputs across varied scenarios. Through our experiments, we observed that the approach we developed offers stable performance, reduced complexity, and consistent accuracy when compared with several existing techniques. As a team, we believe the results validate the effectiveness of our methodology while also highlighting areas that can be strengthened. Although the system performs well for the intended application, it still does not reach the level of detail and multimodal capability demonstrated by more advanced generative models in recent literature. Moving forward, we plan to broaden the dataset, incorporate enhanced learning strategies, experiment with lightweight attention or diffusion-based components, and explore retrieval-or context-aware extensions. These improvements would help us further increase the model's robustness, adaptability, and suitability for real-world environments.

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